

Fuel for thought



Nuclear

**Expert insights into the
future of alternative fuels**

Your trusted adviser in alternative
and low carbon maritime fuel





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Preface

The challenge of maritime decarbonisation is not that it is happening, but that it needs to happen so quickly.

The evolution of sail to its heyday of the great tea clippers took centuries, and the transition to coal-fired steam ships led to a transformation of shipping characterised by greater supply chain mobility and speed. The arrival of oil-fired steam followed by diesel engines made further incremental improvements on the shift from sail to mechanical power. The energy transition the maritime industry faces today is distinct from those earlier evolutions. It is not driven solely by technological advances or economics, but by an environmental imperative, increasingly underscored by social pressure, policy and regulatory demands to reduce emissions.

Decisions are often made without commercial certainty, but in the knowledge that government policy and regulations will push forward change much like Winston Churchill's mandate for the Royal Navy to switch from coal to oil or the mandate of double hull tankers following the grounding of *Exxon Valdez*. In this context, shipowners, charterers, insurers, financial markets and technology suppliers are seeking a better understanding of where the industry is heading. Lloyd's Register (LR) is committed to providing trusted advice and to leading the maritime industry safely and sustainably through the energy transition. Our new Fuel for Thought series puts decarbonisation options under the spotlight, analysing policy developments, market trends, supply and demand mechanics and safety implications. Each edition focuses on a specific fuel or technology, creating a reference point for the industry to overcome upcoming challenges as it faces the next great shift in powering ships.



This edition of Fuel for Thought focuses on nuclear energy, a power source which has had limited adoption in the maritime sector, largely confined to naval applications and Russian icebreaking. With the promise of zero emissions at the point of use and a new, more advanced generation of reactor technologies, nuclear energy is gathering more attention as a zero carbon power source to meet operational and regulatory requirements across a range of maritime applications.

Nuclear energy offers the potential to be as transformational to shipping as the shift from wood to iron, the shift from sail to steam, or the advent of containerisation.



1.1

Chapter 1: Introduction

Introduction from Dr. Mamdouh el-Shanawany, Chair of the Nuclear Energy Maritime Organization (NEMO) and Member of International Atomic Energy Agency's (IAEA) International Nuclear Safety Advisory Group to the Director General (INSAG)

Deploying nuclear energy in the maritime sector holds the key to solving shipping's decarbonisation challenge.

The scale of the challenge presented by the revision of the Greenhouse Gas (GHG) strategy by the International Maritime Organization (IMO) has put nuclear firmly on the agenda. Nuclear propulsion has been used for seven decades by Navies and a few state-owned cargo ships and icebreakers, with an unparalleled safety record. Commercial use has yet to be realised, but advanced nuclear technology is being developed that will be suitable for safe deployment on a new generation of large, efficient cargo ships that can sail at higher speeds with zero emissions.

Shipping's energy transition will take time. Nuclear power is the ultimate decarbonisation solution, but won't be suitable for all ships. Dominant solutions under discussion to decarbonise shipping by 2050, apart from further improvements in energy efficiency, include production of hydrogen and hydrogen-based synthetic fuels, and carbon-capture. These are energy-intensive activities, meaning meeting the IMO's net-zero target this way will require much, much more energy than global shipping consumes today. That energy must be produced without contributing to overall GHG emissions, either from renewables or nuclear. Nuclear has

advantages over renewables as it provides a steady and reliable energy output, and the smallest geographic footprint. Floating nuclear power plants (FNPPs), situated at maritime 'green corridor' hubs, could produce synthetic fuels for shipping using only seawater and air as raw materials.

Nuclear-powered ships and FNPPs producing synthetic fuels are logical solutions to shipping's decarbonisation challenge. To facilitate international movement of commercial nuclear-powered ships, and deployment of mobile FNPPs, both IMO and the International Atomic Energy Agency (IAEA) need to revisit and adjust existing requirements.

By bringing together stakeholders with relevant expertise, NEMO aims to assist nuclear and maritime regulators in the development of appropriate standards and rules for the deployment, operation and decommissioning of floating nuclear power. NEMO will provide expert guidance and promote the highest safety, security, and environmental standards to help unlock the potential of this nascent industrial sector.

LR is a founding member of NEMO, an international membership organisation comprised of maritime and nuclear companies with an interest in floating nuclear power.



1.2

Nuclear fact file



What is nuclear?

Nuclear reactors harness the thermal energy released through controlled nuclear fission to generate heat which can be converted to a mix of electrical, mechanical, or thermal power.

For maritime applications, nuclear reactors generate power for propulsion, industrial mission, and other shipboard needs. Several characteristics of nuclear power promise to bring new operational and commercial models to the shipping industry: Nuclear power can be transformational much in the same way sail gave way to steam and coal gave way to oil. These characteristics include no direct GHG or other emissions, refuelling periods measured in years or decades, high reliability, and limited maintenance while in an operating mode.

Nuclear energy sources are radioactive heavy materials with high energy density; they are strictly regulated across production, distribution, handling, and use. The nature and cost of the fuel lends itself to be capitalised instead of being treated as an operating expense.

Nuclear energy sources vary physically in size, shape, and form depending on reactor design and include solid pellets, rods, and liquids. Fuels vary in their concentration of the fissile isotope U-235 from low enriched fuels, which are tightly regulated, but accessible to licenced private companies, through to the highly enriched fuels used in military or research reactors in only a handful of countries.

The cost structure and regulatory controls around operating nuclear reactors will bring new relationships between ship operators and reactor owners. Certain ship owners and operators may buy the power generated by a reactor rather than owning the reactor itself, separating the shipowner from the complexities of licensing and operating nuclear reactors.

A key operational differentiator for nuclear reactors is that they do not require bunkering. Instead of regularly taking on fuel for the voyage ahead, nuclear reactors have the potential to run for years or even decades without refuelling. Refuelling schedules are expected to range from a minimum of five to eight years and potentially as long as thirty years.

Nuclear Fission

The fission of heavy atomic nuclei results in several lighter elements and released energy. A nuclear reactor safely sustains nuclear fission and captures the energy it creates as heat which is then used to generate a mix of electrical, mechanical, or thermal energy power.

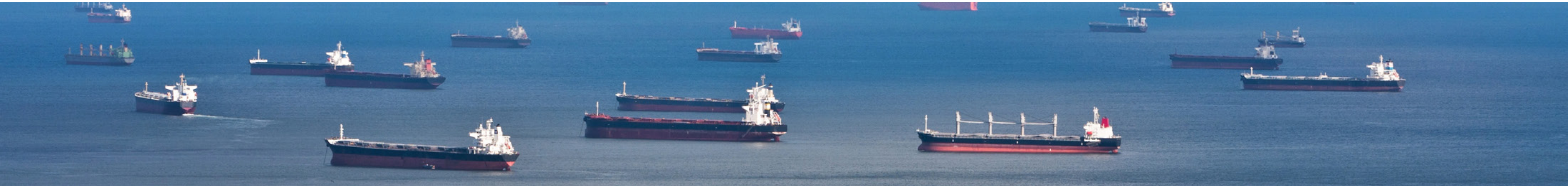
Properties table

 No GHG emissions	 No sulphur (SOx)
 No nitrogen (NOx)	 Energy density comparison Uranium 235 – 3,900,000 MJ/kg Diesel – 45 MJ/Kg

Advantages and disadvantages of Nuclear

The following table offers a brief insight into the benefits of nuclear power for shipping and its challenges

Advantages and potential	Challenges and issues
Excellent safety record	Public perception and the social licence to operate
Infrequent refuelling required (bunkering)	Regulation needs updating and is more involved than for conventional ships
Zero emissions	New technologies unproven in maritime
Low onshore infrastructure requirements	High upfront capital investment for reactor owners
Straightforward onboard requirements	Disposal of spent fuel and radioactive material
Much of the depleted fuel can be used for power using new-generation nuclear technologies.	Movement of nuclear material



1.3

Nuclear readiness as a marine energy source

LR has collaborated with industry stakeholders to build a comprehensive assessment of nuclear fuel production and supply, and the reactor technologies under development for onboard power generation.

LR's Maritime Decarbonisation Hub has developed a framework to measure the current readiness of multiple aspects of several fuels in its [Zero-Carbon Fuel Monitor publication](#).

A lot of focus is often put on the technology readiness level (TRL) of new solutions, which assesses the maturity of solutions to become marine application ready. TRL addresses the question of how close the technology is to being proven, scalable and safe. However, technology readiness is just one element in assessing overall solution readiness for commercial maritime adoption. Investment readiness level (IRL) evaluates the commercial maturity for marine solutions, considering the financial proposition, industry, supply chain dynamics and market opportunities. Community readiness level (CRL) gauges the societal maturity of the marine solution, considering its acceptability and adoption by both individuals and organisations; it encompasses regulatory, sustainability and community acceptance aspects.

TRL is assessed on a scale of one to nine, IRL and CRL are on a scale of one to six.

LR's Maritime Decarbonisation Hub uses the outputs of the monitor to identify research, development and deployment projects that will advance solution readiness and accelerate a safe and sustainable transition to net-zero GHG emissions.

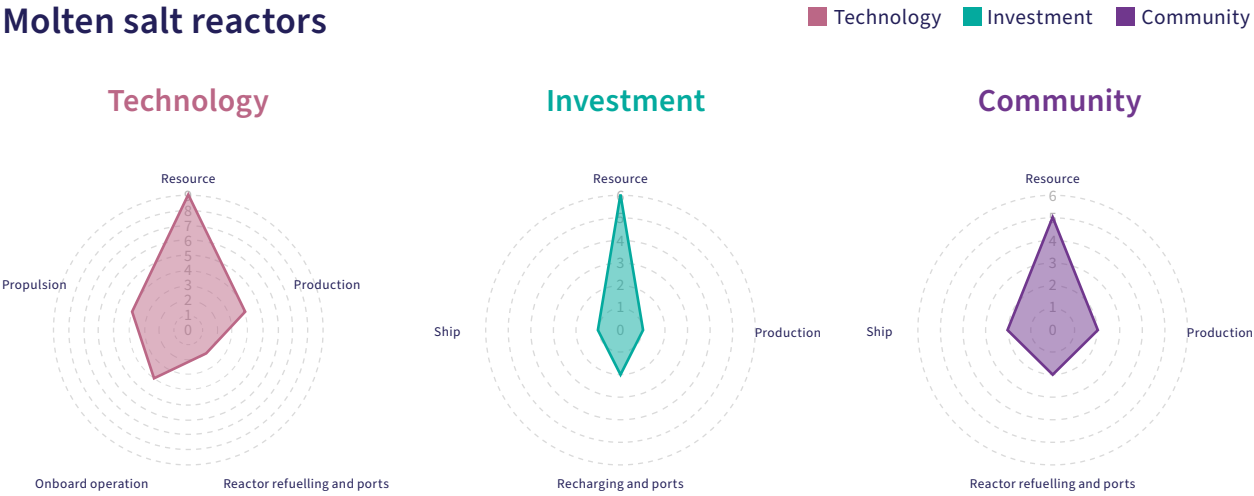
There are nuclear vessels in operation in military and icebreaking operations, many using Highly Enriched Uranium (HEU) in reactor types not suitable for commercial use. The readiness assessments for nuclear power in shipping reflect the work needed for nuclear power solutions to be ready for adoption in commercial maritime operations. The new generation of advanced nuclear reactors are being developed for applications on land as well as at sea, inviting a broader investor base for advancing the fundamental technologies and reactor designs.

Obtaining the social licence to operate for nuclear power is a prominent challenge due to public perceptions regarding safety.

The specifics of individual reactor technologies are explored in greater detail in Chapter 5. Definitions of the IRL, TRL and CRL levels can be found in Annex 1.

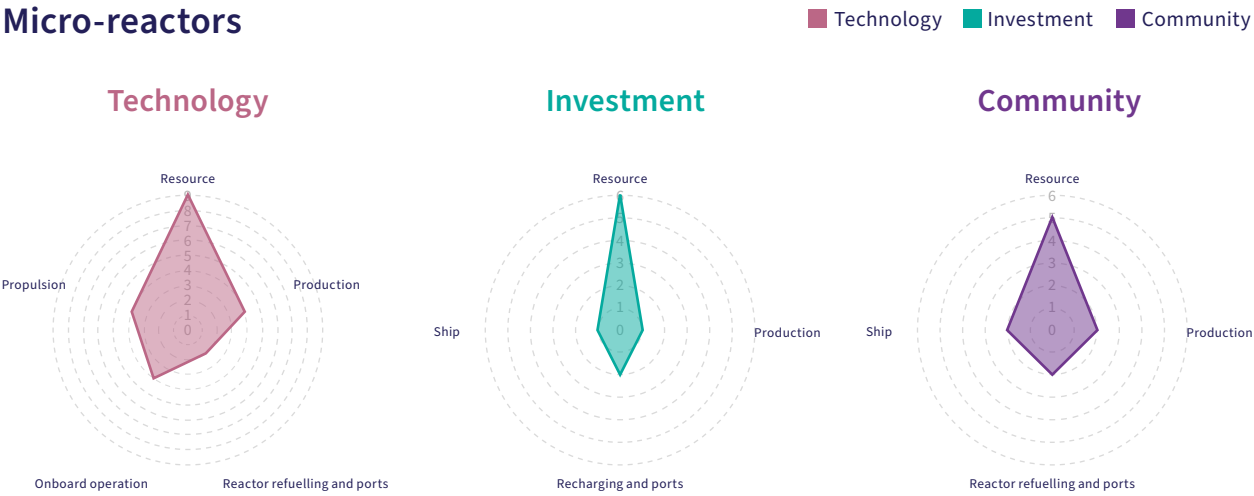


Molten salt reactors



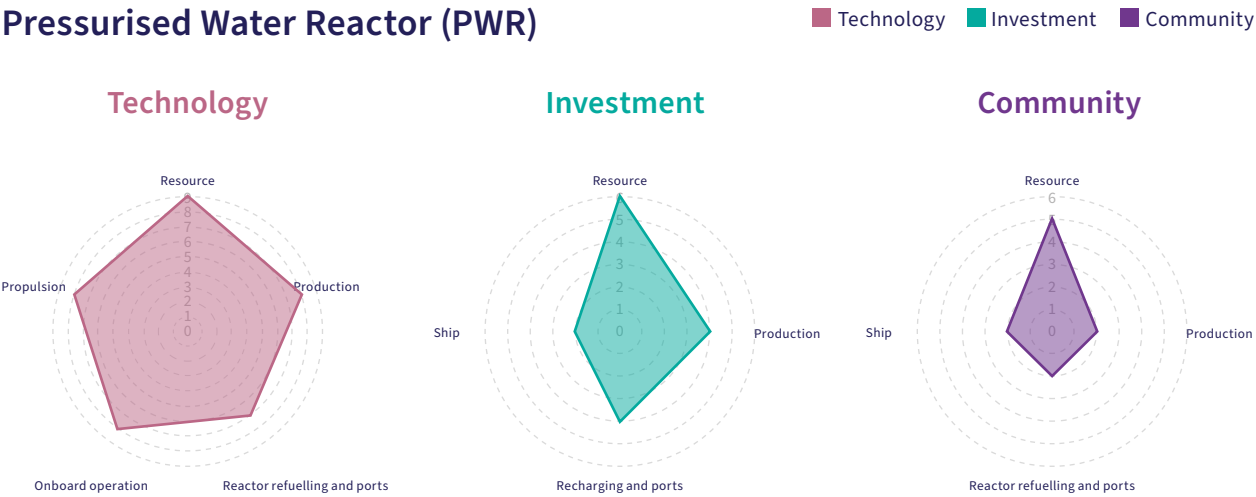
Technology Readiness Levels (1–9), Investment and Community Readiness Levels (1–6)

Micro-reactors



Technology Readiness Levels (1–9), Investment and Community Readiness Levels (1–6)

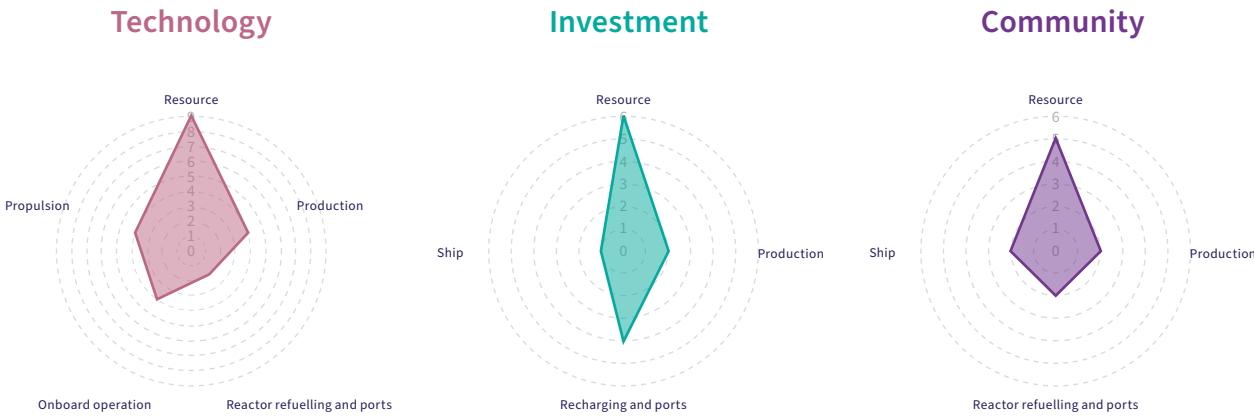
Pressurised Water Reactor (PWR)



Technology Readiness Levels (1–9), Investment and Community Readiness Levels (1–6)

High temperature gas reactors

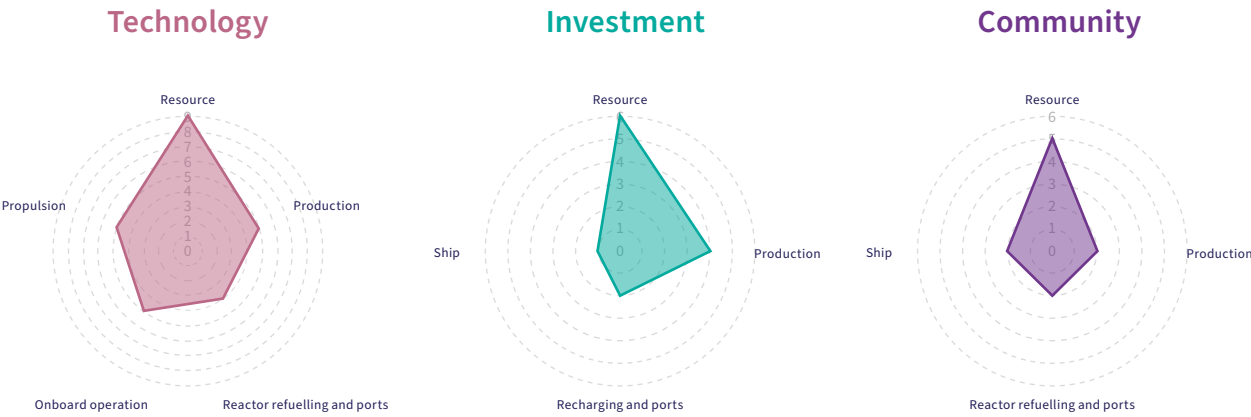
Technology Investment Community



Technology Readiness Levels (1–9), Investment and Community Readiness Levels (1–6)

Liquid metal cooled reactors

Technology Investment Community



Technology Readiness Levels (1–9), Investment and Community Readiness Levels (1–6)



2.1

Chapter 2:

General Safety and radioactivity issues

Public perception of nuclear technologies

Nuclear energy has a unique challenge among the alternative fuels for the maritime industry in the area of public perception. Despite the strong safety record of the nuclear energy sector, events like Fukushima and Chernobyl continue to influence public opinion because of the potential severity of nuclear accidents.

The primary public concern around nuclear technologies is the effect on humans of high levels of radiation that increase the probability of cancer. Low levels of radiation are being better understood through study as it is unknown whether low level radiation poses a threat to human health. To earn the social licence to operate, it is expected that new applications will have to thoroughly demonstrate that radiation exposure during normal or upset events is no different than in everyday life. The public demands that inherently safe design is expected to be the minimum for nuclear systems.

Robust regulations are in place to limit human exposure to radiation both during normal operation of nuclear facilities and in emergency situations, and are explored in this chapter.

Another public concern is the potential for a reactor meltdown that results in excessive released radiation. New reactor designs have various passive safety features to prevent the release of radioactive materials and minimise the impact of a containment breach should one occur.

Another major concern is the handling of radioactive waste, including spent nuclear fuel and contaminated materials. To earn the social licence to operate, the eventual disposal of spent nuclear material must be well planned and suitable to prevent persistent problems. International standards for the safe storage and disposal of such material are already in place and covered in chapter 4.3.

Concerns regarding bad actors using mobile nuclear reactors for nefarious activity are addressed by thoroughly demonstrating compliance with the “security and safeguarding” aspects of the IAEA “3S” requirements.

Public support for shore-based nuclear power is increasing as the need for secure, reliable, and low emission power becomes more important. Public engagement will be an important part of the commercial adoption of the technology.



General safety and radioactivity

Introduction

The nuclear energy sources used in reactors are highly radioactive materials; exposure to them is detrimental to human health. For this reason, multiple safety systems are put in place to shield workers from any radioactive material, and acceptable exposure limits for those working at nuclear sites are set conservatively low.

The health impacts of exposure to ionising radiation may not be immediately apparent and vary depending on the dose received. Acute radiation poisoning can lead to death within days. Exposure to elevated ionising radiation levels increases the risk of cancer.

Radioactivity cannot be detected by human senses, creating a risk of exposure without knowledge. There are tools to measure radioactivity, track exposure to radiation, and alert when radiation levels rise.

The stringent technical and operational safety regulations around nuclear power generation reflect the potential risks of radiation exposure and environmental contamination.

Exposure to radiation

We are all exposed to radiation during the course of our daily lives from natural sources such as cosmic radiation from space, and smaller amounts from man-made sources. This background radiation varies geographically depending on local geology, altitude, and the built environment. Medical procedures are another common source of radiation exposure.

According to the US National Council on Radiation Protection and Measurements (NCRP), the average US citizen is exposed to 6.2 mSv annually, comprising around 50% background radiation, 48% exposure from medical procedures, <0.1% occupational exposure, <0.1% industrial exposure, and 2% consumer exposure.

IAEA give recommendations that any exposure above natural background radiation should be kept as low as reasonably practicable, but below the individual dose limits (IDL). The IDL for radiation workers averaged over 5 years is 100 mSv, and for members of the general public, is 1 mSv per year. There are variations in measurement and application of IDL between nations.

The IAEA reports that annual doses received by radiation workers are found to be considerably lower than IDL.





Reactor safety

The designs of nuclear reactors under consideration for use in the maritime industry have specific passive safety features in place to prevent nuclear accidents. One weakness of older generation reactor designs is their reliance on external cooling systems to prevent the fuel overheating. This cooling is required even when the reactor is shut down. In these older designs, a prolonged failure of the cooling systems such as a loss of power to pumps could lead to excessively high temperatures in the reactor.

The latest reactor designs include passive safety features such as cooling systems which do not rely on emergency generators and pumps for safety-related functions, making them ‘walk away safe’ in the event of a malfunction. Passive core cooling systems reduce the risk of nuclear accidents and their consequences.

In Molten Salt Reactors (MSR), for example, the non-pressurised nature of the reactor limits dispersion of any radioactive material in the event of a failure. MSR designs use various means to initiate passive shutdown in the event of a temperature rise by exploiting the inherent characteristics of the molten salt.

Nuclear security

Anti-terrorism considerations are a part of the regulatory evaluation and approval process for reactor designs and the development of emergency planning zones and contingency plans. The same passive safety features that minimise the risk and consequences of malfunction in next generation reactor designs also reduce the likelihood and consequences of containment failure in the event of deliberate attack.

It is not possible for a commercial nuclear reactor to explode like a nuclear weapon. The enrichment level of nuclear energy sources in civilian applications is limited to prevent their use in the creation of nuclear weaponry. By keeping enrichment below 20%, the fuel in commercial reactors remains unattractive as a feedstock for creating weapons grade uranium due to the technology and effort required for further enrichment.

Nuclear reactor modules have to conform to the IAEA requirements for Safety, Security, and Safeguarding (IAEA 3S's) with which conventional power sources do not have to comply. This makes the nuclear powered vessel more resistant to bad actors than conventional vessels.

2.3

Nuclear regulation

The nuclear energy sector is tightly regulated and has one of the best safety records among electricity generation methods, far exceeding fossil fuel power sources and on a par with wind and solar.

The industry is overseen by national nuclear regulators such as the UK Office for Nuclear Regulation (ONR), Korea's Institute of Nuclear Safety (KINS), and the US Nuclear Regulatory Commission (NRC). The IAEA promotes and supports the establishment of comprehensive regulatory frameworks to ensure the safety of nuclear installations throughout their lifetime. These regulatory frameworks consist of relevant legislation, regulations, guidance and a robust leadership and management programme for safety.

IAEA regulations for land-based nuclear reactors are well-established. These rules are under review to prepare for the particular requirements and implications of Small Modular Reactors (SMRs) in mobile or transportable applications and as serial manufactured products. Importantly for the maritime industry, work is also underway to assess the viability of floating nuclear power plants under existing rules, a process with significant relevance for the future development of rules for nuclear ships at IMO and IAEA.

Nuclear regulation in shipping

Chapter VIII of the International Convention for the Safety of Life at Sea (SOLAS) gives the basic requirements for ships provided with a nuclear power plant, creating a framework encompassing design, construction, operation, maintenance, surveying, salvage, and decommissioning of nuclear reactors on ships, including as a means of propulsion.

IMO A.491 Code of Safety for Nuclear Merchant Ships was adopted in 1981 to supplement SOLAS Chapter VIII, adding further guidance on the safety criteria for nuclear ships. The code accompanied the 1962 Brussels Convention on the Liability of Operators of Nuclear Ships which has not yet entered into force and is unlikely to enter into force.

Developed on the basis of reactor technologies and safety frameworks of the 1970s, Chapter VIII and the Code are in need of a thorough review considering the technological developments of the past four decades, the ongoing developments in reactor design, and the evolution of safety management systems and quality systems.

LR provided a gap analysis on maritime safety and liability for a wider gap analysis on the Code of Safety for Nuclear Merchant Ships submitted to the IMO Maritime Safety Committee's 108th session by the World Nuclear Transport Institute (WNTI). The document identifies areas of the code in need of revision to guide the design and safety assessment of nuclear merchant ships.

The development of specific regulations for nuclear reactors used in shipping will require the close cooperation of the International Maritime Organization and IAEA due to their respective responsibilities regulating shipping and the peaceful use of nuclear energy.

The UK Merchant Shipping (Nuclear Ships) Regulations

The Merchant Shipping (Nuclear Ships) Regulations 2022 came into force late in 2022 alongside a Marine Guidance Note on nuclear ships, MGN 679 (M).

The rules effectively transpose SOLAS Chapter VIII and the nuclear code into UK law, and as such it is limited to considering pressurised water reactors. The regulation and MGN give an idea of the role national governments may expect to play in the adoption of nuclear powered shipping.

MGN 679 states: "Although it is not possible at the present time to provide extensive guidance on this subject, the Maritime and Coastguard agency (MCA) will provide clarification on a case-by-case basis. The MCA will also endeavour to provide additional guidance when further experience of nuclear-powered ships is acquired."

Some notable contents include:

- The nuclear reactor installation for a United Kingdom nuclear ship must be approved by the MCA before construction of the ship commences.
- The Nuclear Code is based on pressurised light water type reactors. Other types of reactors will require special considerations and may be approved by the MCA on a case-by-case basis.
- The safety assessment and the voyage plan of a nuclear-powered ship must have been notified to the MCA at least 12 months before the arrival of the ship in United Kingdom waters, including a United Kingdom port.
- A nuclear-powered ship should be designed, constructed, tested, inspected, operated and decommissioned under a Quality Assurance Programme (QAP) and at all stages in the ship's life cycle, there should be a single organisation responsible for the management and control of the overall QAP.
- A fully detailed operating manual should be prepared and continuously updated for the information and guidance of the operating personnel in their duties on all the matters relating to the operation of the nuclear power plant with a particular attention to safety.
- Where a nuclear ship is at a fixed-point mooring or alongside a berth, and where there is work on the ship involving ionising radiation, the Radiation (Emergency Preparedness and Public Information) Regulations 2019 apply.



Emergency Planning Zones

On shore, nuclear facilities are surrounded by Emergency Planning Zones (EPZ) within which plans must be in place to respond to nuclear accidents to protect human health and the environment.

The current Pressurised Water Reactors used in naval vessels can have large EPZs when in port spanning multiple kilometres, which create large liabilities for the vessel operators; the vessel is mobile and so its EPZ can encompass significant assets and infrastructure on land and on water. For nuclear naval vessels, these liabilities are underwritten by their governments to enable operation.

Nuclear ships are specifically not covered by The Convention on Limitation of Liability for Maritime Claims 1976. This means liability falls to the operator of a shipboard nuclear power plant.

For risks to be underwritten on the commercial insurance market, the EPZ of a nuclear ship needs to be shrunk to the boundary of the vessel, a process which may be enabled by the safety characteristics of certain SMR technologies. There is a broad expectation that safely reducing the EPZ to the vessel boundary is achievable for SMRs.

There is precedent for the reduction of EPZ size for new generation reactor technologies, including a methodology by NuScale which was validated by the US Nuclear Regulatory Commission. The methodology can be used to develop EPZs at the size of a power plant's site boundary on land.

Risk-based certification

While not a solution in itself, risk-based certification is an approach that can be employed to adapt regulations to be fit for current and approaching nuclear technologies in shipping. The LR Risk Based Certification requirements follow an approach where the fundamental requirement is to demonstrate an equivalent level of safety to that achieved with conventional oil-fuelled systems as well as the general expectations of the IAEA 3S requirements.

Safety regulations for nuclear ships are in need of update but will likely remain based on risk assessments and risk-based certification needs, and therefore high level and more goal-based in nature than prescriptive. International maritime safety regulation on nuclear fuel-handling will require discussion at the IMO and within International Association of Classification Societies (IACS) and other safety forums, with classification issuing guidance on fuel system design, handling and other critical safety considerations. The approval process is outlined in the IMO Guidelines for Approval of Alternatives and Equivalents (MSC.1/ Circ.1455).

LR has developed a Risk-Based Certification (RBC) process which is consistent with and upon MSC.1/Circ.1455 and other related IMO guidelines, yet equally applies to non-SOLAS projects. RBC is used where risk assessment is required to inform certification and provide confidence in new, novel and alternative designs. For an alternative fuel project, the risk-based process needs to meet the mandatory requirements in SOLAS Reg.II-1/55, the guidance in MSC.1/Circ.1455, and be undertaken in accordance with the LR RBC process.

2.4

Specific refuelling considerations

Refuelling is an area where nuclear power presents the potential for significant operational changes for the maritime industry. The high energy density of nuclear fuel creates the opportunity for reactors that will operate for multiple years before they need refuelling or replacing. Some designs may provide power for the entire lifespan of a ship.

In the existing fleet, Russian nuclear icebreakers are refuelled every five to seven years; some nuclear submarines are designed so that their reactors never need refuelling over their 30-35 year operational lifespan.

Whereas bunkering traditional and alternative fuels – or recharging the batteries of electric vessels – is the responsibility of the crew, the refuelling of a nuclear reactor will be a specialist operation carried out by trained experts at a specialist facility. Depending on future reactor designs, refuelling may involve replacing fuel in the reactor, topping up the reactor's fuel, or taking advantage of the modular nature of some designs and replacing the reactor entirely.

Operational experience in refuelling nuclear reactors for ships is largely confined to military applications where refuelling often coincides with a mid-life refresh of a vessel's systems.

Fuel costs

Nuclear energy sources contain the power to meet a vessel's propulsion and other energy needs for multiple years. The price of

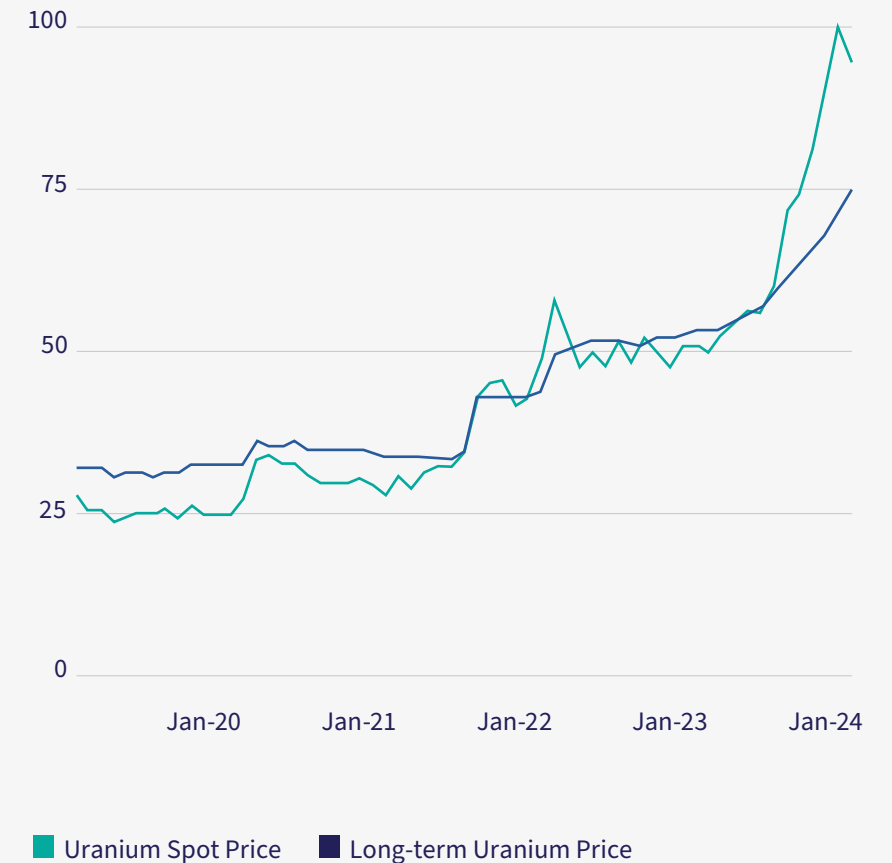
fuelling a reactor will be much higher than filling a ship's bunker tanks with traditional fuels once, but becomes cost competitive when compared to a ship's total bunker expenses over the much longer operational periods a reactor fuelling sustains.

Because of the high cost of buying multiple years of a ship's energy requirements upfront and the tight regulation of the distribution of nuclear energy sources, it is not expected that ship owners and operators will purchase this fuel as an operational expense. Instead, the maritime industry may lease a reactor and its operation and contract for energy output from onboard reactors. The fuel itself can be treated as a capital expense from the outset. In this scenario, the relevant operational cost is the agreed price per Megawatt hour (MWh) of energy on the ship rather than the cost of refuelling the reactor.

Uranium prices are tracked as a commodity in US dollars per weight of U3O8 equivalent. The market price of Uranium has tracked upwards since 2021 driven by a recovery from prolonged lower demand in the wake of Fukushima, a growing acceptance of the role of nuclear power in decarbonisation, disruption in some supplier nations, and growing geopolitical tensions.

Uranium price trends

(USD/lb of U3O8 equivalent)

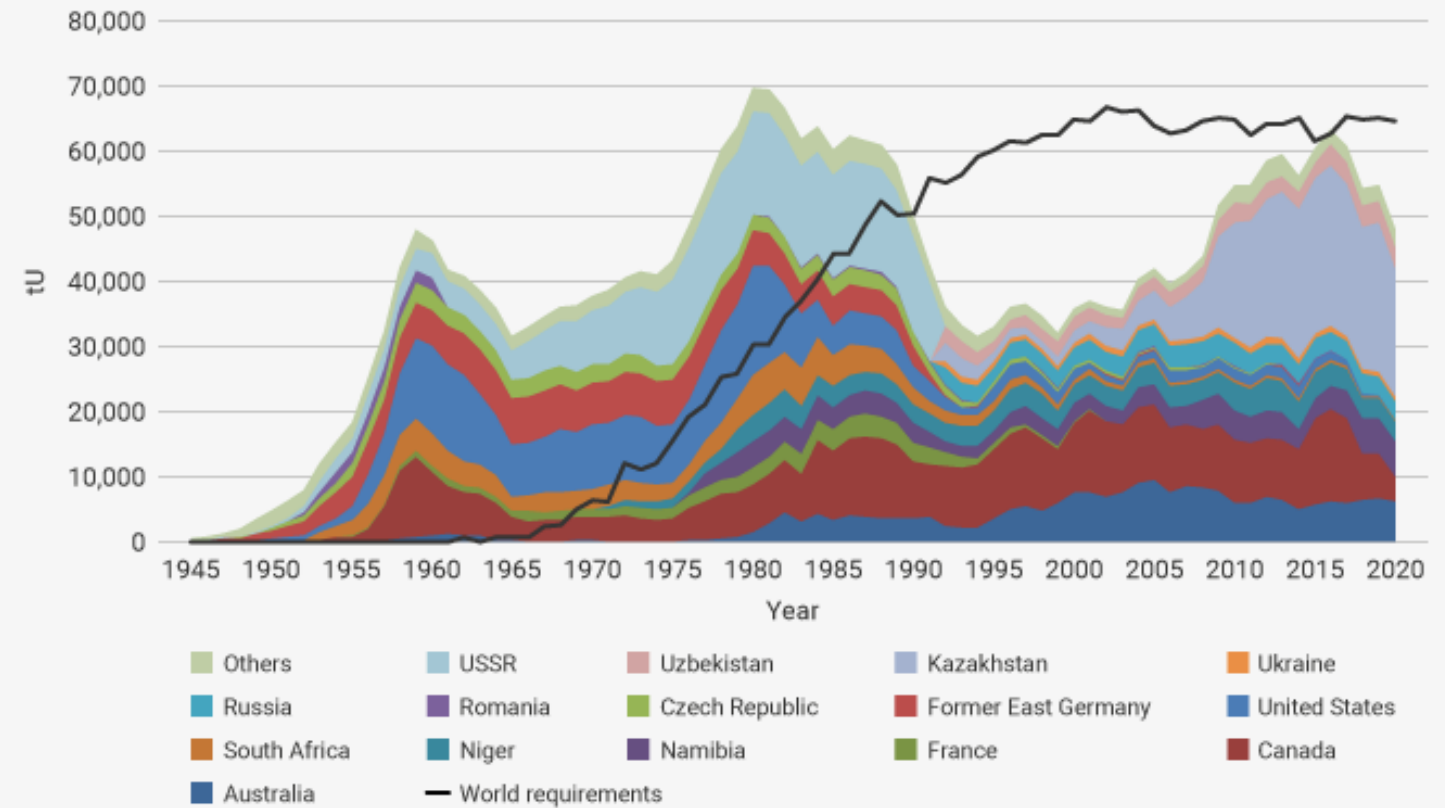


Source: <https://www.cameco.com/invest/markets/uranium-price>

World uranium production

Kazakhstan, Canada and Namibia lead the world in uranium production from mines. Global production volumes have slowly increased since a low in 2020 and are expected to continue to grow in the 2025-2040 period alongside demand for nuclear power. Uranium is more abundant than Tin and Zinc. The US, Japan and China have been working on a process to extract Uranium from seawater; when extraction techniques become economically viable and efficient, uranium from seawater could last for thousands of years owing to the suggested 4.5 billion tonnes of Uranium in sea water.

Uranium production and demand



Sources: OECD-NEA/IAEA, World Nuclear Association

2.5

Port readiness and regulation

Work is underway to create regulations and risk frameworks to enable nuclear-powered commercial vessels to call at commercial ports. While this work is ongoing, nuclear-powered naval vessels regularly receive permission to visit civilian ports and have done so for decades without incident.

There are also historical precedents of nuclear powered merchant vessels sailing internationally and calling at multiple ports.

The movement of nuclear ships between ports is enabled by the flag state or licence holder's national regulator providing the reactor licence and regulating the reactor and vessel. Agreements are put in place between nations to recognise those licences in order to allow port entry. Under the UK Merchant Shipping (Nuclear Ships) Regulations 2022, for example, the safety assessment and the voyage plan of a nuclear-powered ship must be notified to the Maritime and Coastguard Agency at least 12 months before the arrival of the ship in UK waters, including a UK port.

The first nuclear powered merchant ship *NS Savannah* was built by the US government and operated for 10 years between 1962 and 1972. The vessel toured multiple ports in the US and around the world to promote the peaceful use of nuclear power as a centrepiece for President Eisenhower's Atoms for Peace program. *NS Savannah* visited 32 US ports and another 45 ports in 26 countries, including transiting the Panama Canal. *NS Savannah* was however excluded from some ports, including in Australia, and New Zealand.

The German nuclear-powered research vessel *Otto Hahn* visited 33 ports in 22 countries during its 10 years of operation from 1969. Notably the ship was not permitted to pass through the Suez Canal.

Infrastructure requirements

Compared to fuel oil and the other alternative fuels, nuclear vessels require very little landside infrastructure to support their routine operations. By removing the need for bunkering, nuclear-powered ships do not need to rely on the pipes, tanks, bunkering barges, cryogenic storage and other facilities in port that make up the bunkering infrastructure.

It is expected that designated ports will specialise in the provision of maintenance services for nuclear vessels and that these services will be required much less frequently than regular bunkering.



2.6

Fuel quality

The production of nuclear energy sources is tightly regulated. Extensive quality controls ensure that fuels are manufactured to precise standards. Short supply chains and sealed transportation flasks for nuclear fuels reduce the risk of fuel contamination to near zero.

There are multiple international producers of nuclear energy sources for reactors. The market is large enough to create competition but kept small by the high barriers to entry including cost and regulation.

There are many different types of nuclear energy sources which can be arranged by the concentration of the fissile isotope U-235. Natural uranium is around 0.7% U-235, Low Enriched Uranium fuels are below 20% U-235, and Highly Enriched Uranium contains 20% or more U-235. Civilian reactors, such as those on commercial ships, will only be allowed to use low enriched fuels.

For many of the advanced reactor designs under consideration for future use in merchant shipping, the energy source would be High-Assay Low-Enriched Uranium (HALEU) with enrichment between 5% and 20%.

Quality Management Systems for the supply chain of the nuclear energy sector are in place under ISO 19443:2018 and due to be replaced by ISO/AWI 19443.

As with fuel cost, fuel quality is not expected to be a concern for the shipowner but for the reactor owner.

2.7

Summary

Nuclear power generation has a strong track record in safety and an engaged set of regulatory bodies overseeing the safe development of new reactor technologies. The introduction of nuclear powered vessels to the maritime industry at any scale will require widespread updates to regulations, including SOLAS Chapter VIII. Cooperation between the IMO and IAEA will be necessary to create a harmonised regulatory system for nuclear powered ships.

The shipping industry has relatively limited experience with the operation of nuclear powered merchant ships; most marine experience is with naval vessels that use older reactor types which are not suitable for commercial shipping.



3.1

Chapter 3: Drivers for Nuclear

Regulations

The following discussion focuses on various regulatory drivers behind the interest in nuclear propulsion in shipping. For safety regulations, see Chapter 2 of this report. The regulatory drivers for nuclear powered shipping are the same as those for other alternative fuels being a reduction in emissions. Nuclear power has the advantage of zero operational emissions, creating a direct path to compliance with regulations aimed at reducing GHG emissions from shipping and the ultimate goal of a net zero emissions industry.

EU Regulations

Some of the most mature emissions regulations for shipping are from the European Union (EU). A recent signal of the EU's stance on nuclear power was the inclusion of "advanced technologies to produce energy from nuclear processes" and small modular reactors as net-zero technologies under the Net-Zero Industry Act (NZIA). However, the bloc fell short of including nuclear as a

"strategic technology", which would have brought support to projects in permitting and public procurement.

Shipping companies need to be aware of five elements of the EU Fit for 55 package that impact shipping. The Fit for 55 package is the bloc's overarching decarbonisation strategy across society and business. It includes:

- **A revised regulation on the Monitoring, reporting, and verification of greenhouse gas emissions from maritime transport regulation (EU MRV)**
- **A revised Directive on the EU emissions trading system (EU ETS)**
- **A new FuelEU Maritime Regulation**
- **A revised Alternative Fuels Infrastructure Regulation (AFIR)**
- **A revised Renewable Energy Directive (RED III)**

[Analysis from LR highlights](#) how these interlocking requirements will drive ship owners to adopt more stringent vessel efficiency strategies, as well as new low-carbon fuels. Tank-to-wake (TtW) CO₂ emissions from cargo and passenger ships of 5,000GT and above, reported under the MRV system in 2024, will be subject to EU ETS in 2025.

For general cargo ships of 400GT to 5000GT, and for offshore ships of 5000GT and above, MRV reporting will be applicable from 2025.

Under EU ETS, shipping companies with responsibility for ships within its scope will need to buy allowances to cover greenhouse gas (GHG) emissions for intra-EU voyages, at-berth emissions, and for half of the GHG emissions released during voyages to and from the EEA. GHG emissions to be surrendered currently include CO₂ emissions but, from 2026, will also encompass NO₂ and CH₄ emissions on a CO₂ equivalent basis.

There are no free allowances as there were for other sectors in early stages of the EU ETS, but for shipping there will be a phase-in period where shipping companies will have to hand in allowances that cover only a percentage of the verified emissions for a particular year (see infographic on right).

For EU ETS, a review by December 2026 will consider the addition of offshore ships of 400GT to 5000GT.



Settlement of allowances for each year will be required by 30 September of the following year.

Fuel EU Maritime

This regulation supplements the ETS and promotes the use of alternative fuels accounting for the GHG intensity of energy used onboard on a well to wake basis. It creates an incentive to use lower or zero-emission fuels, noting that carbon pricing and policies that support energy efficiency will not be sufficient on their own to meet EU targets for decarbonisation across the bloc. From 2025, shipping companies are required to meet stepped reductions in the GHG intensity of energy used onboard, as shown in the chart to the right. In addition, FuelEU sets a requirement to have zero at-berth emissions for container and passenger ships, coming into effect from 2030.

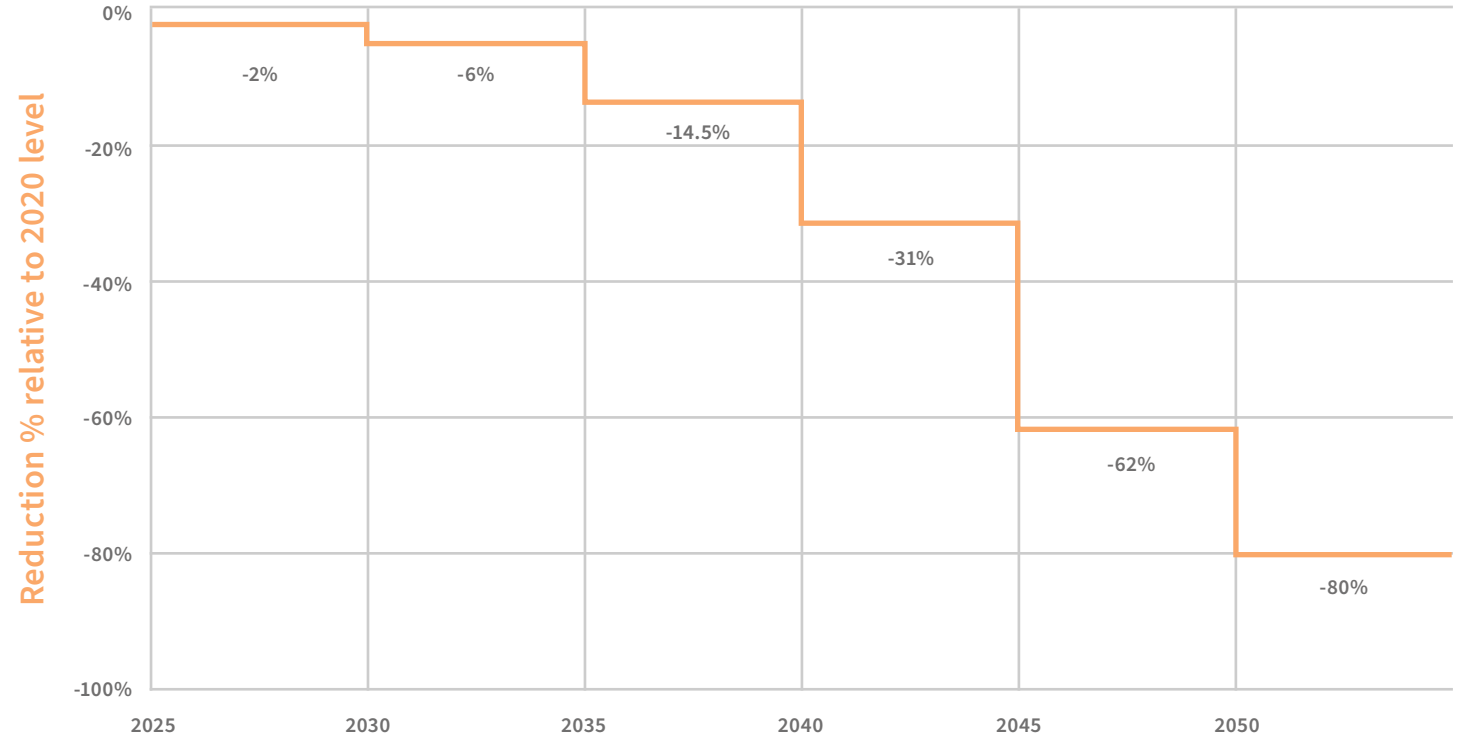
[The FuelEU Maritime Regulation](#) requires submission of a new monitoring plan to verifiers. Assessment for each ship should indicate the chosen method used to monitor and report the amount, type and emission factor of energy used on board. From 1 January 2025, each ship's energy consumption information needs to be recorded and collected. The full year's data will then be submitted for verification by 30 March of the following year with penalties due for non-compliance with that year's GHG intensity reduction target. A review of FuelEU by December 2027, and every five years thereafter, may increase its scope.

Nuclear power is not accounted for in the FuelEU Maritime Regulation, and is absent from the list of zero-emission technologies (Annex III), which is currently limited to fuels cells, on-board electricity storage, and on-board electricity production from wind and solar energy. The regulation does have a mechanism for the addition of new technologies to Annex III "where these new technologies are found equivalent to the technologies listed in that Annex in the light of scientific and technical progress."

Should nuclear power be added to Annex III, it would fit the broader requirements of a zero-emission technology under the regulation as it does not emit:

- carbon dioxide (CO₂)
- methane (CH₄)
- nitrous oxides (N₂O)
- sulphur oxides (SOx)
- nitrogen oxides (NOx)
- particulate matter (PM)

FuelEU Maritime Reduction Factor



Above: Reduction in GHG intensity of energy used on board from 2020 levels (%).

Pooling

Included in the provision of each vessel's data is the notification of the decision to pool vessels. Pooling allows owners and pool managers to bring together vessels that have been operated within a fleet, within a company or among companies. The objective is to encourage the deployment of new vessels using low- GHG-emission or zero-GHG-emission solutions, instead of focusing only on improving the performance of existing vessels. Pooling allows the reduced GHG intensity of one vessel to be shared amongst a fleet to reduce the GHG intensity of the individual vessels and reduce their exposure to financial penalties which can be incurred under FuelEU Maritime. Additionally, pooling aims to reduce dependence on biofuels and encourage the uptake of alternative low and zero GHG fuels for early adopters which is required to begin to scale up adoption.

As noted in a [recent LR article](#), the ability to pool emissions penalties and surpluses has far-reaching significance. A recent analysis by Core Power showed that for a 12-vessel pool from 2030-2034, around \$463m in in Fuel EU maritime penalties, \$84m in EU ETS costs and \$260m in fuel costs could be saved by replacing a single VLSFO-fuelled container ship with a nuclear-electric equivalent vessel.

GHG emission factors for fuels under Fuel EU Maritime

Different fuels are assigned different emission factors based on their greenhouse gas intensity, although nuclear is not currently included. Methods for determining the greenhouse gas emission factors of all fuels are provided in Annex I to the FuelEU Maritime Regulation.

International regulations (International Maritime Organization)

IMO regulations relating to controlling CO₂ emissions on a global scale are in force and have focused on ship energy efficiency to date. In 2018, following the 2015 Paris Climate Agreement, the IMO agreed an initial GHG strategy to outline a pathway to reduce shipping emissions by focussing on CO₂ reduction from ships, with the aim of keeping the rise in global temperatures to within 1.5 degrees centigrade of pre-industrial temperatures. The initial strategy led to the development of short-term measures, namely the Energy Efficiency Existing Ship Index (EEXI) and the Carbon Intensity Indicator (CII).

At the 80th meeting of its Marine Environment Protection Committee (MEPC 80) in July 2023, the IMO adopted a revised GHG reduction strategy. This aim to achieve net-zero CO₂ equivalent emissions by, or around, 2050 with indicative checkpoints along the way for shipping to aim for, including:

- Reducing total GHG emissions from international shipping by at least 20% by 2030, striving for 30%
- Reducing total GHG emissions from international shipping by at least 70% by 2040, striving for 80%

All reductions are compared to 2008 levels. There is also a target for low- or zero-carbon fuel uptake of at least 5%, striving for 10%, as well as a reduction of carbon intensity of international shipping compared to 2008 levels by at least 40% by 2030.

The revised GHG reduction strategy sets a timeline for the adoption of mid- and long-term measures to reduce emissions from shipping, requiring an agreement on mid-term measures at MEPC 83 in spring 2025 in order for those measures to enter into force in 2027.

The measures will include both a technical, and an economic, element. The IMO has now completed, agreed, and adopted fuel lifecycle analysis guidelines and continues to review them. These will support the technical and economic measures by enabling calculations of well-to-tank emissions (the emissions associated with the production and supply of a marine fuel) as well as well-to-wake emissions (also adding in emissions as a result of the fuel's use on the vessel).

Lifecycle analysis

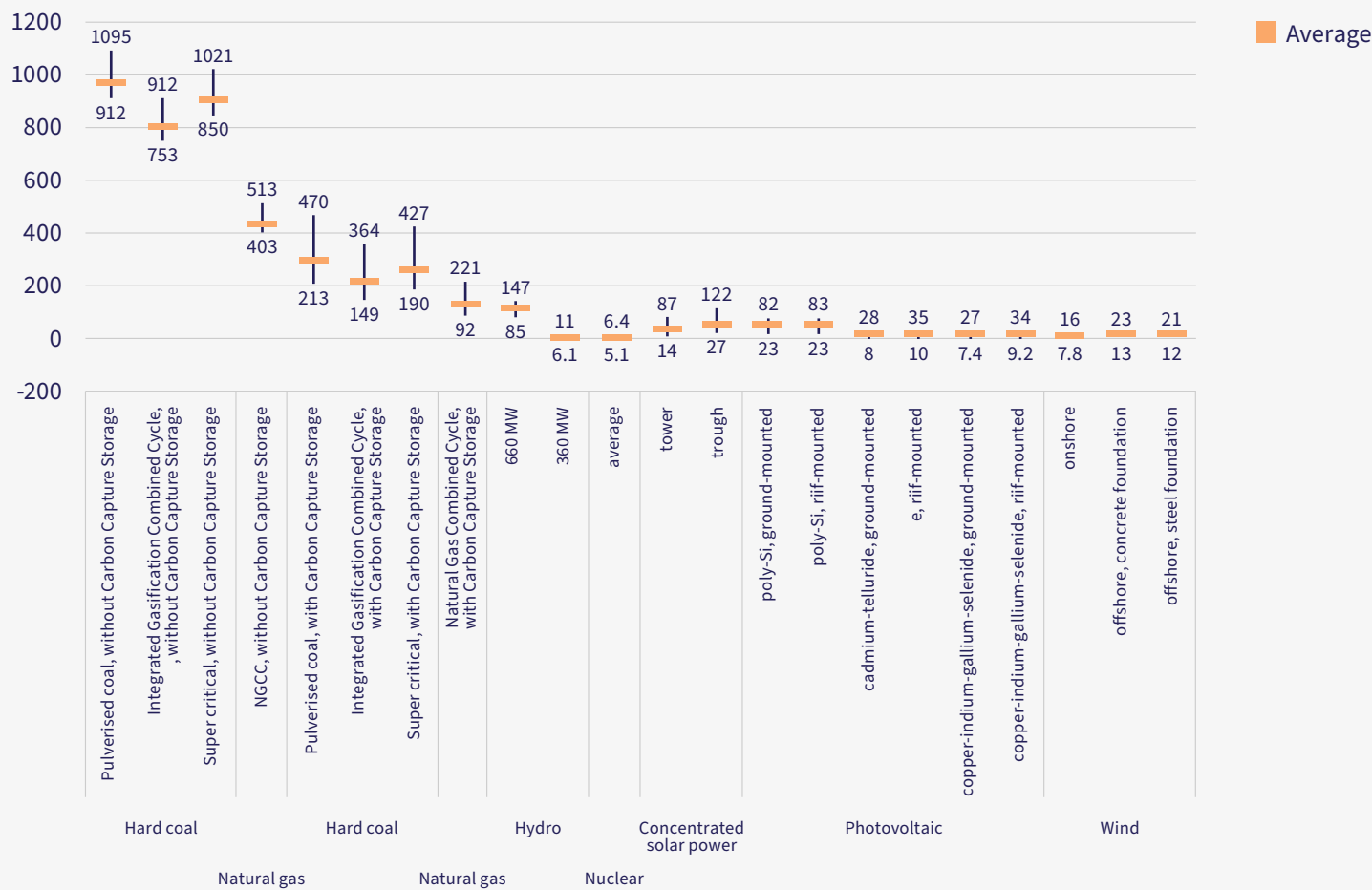
Nuclear power generation does not involve the use of hydrocarbons as a fuel or any form of combustion, and does not require the use of a pilot fuel. The reactor itself has zero operational emissions, but there are indirect life cycle emissions to consider in the mining and refining of nuclear fuels and the construction of reactors. The IMO Lifecycle Analysis guidelines do not include nuclear reactors for shipboard power. The guidelines only refer to nuclear power as an energy source for use in the creation of other fuels such as hydrogen and ammonia.

For an idea of the lifecycle analysis of nuclear power for ships, we can look to shore-based nuclear power generation. The lifecycle emissions for nuclear power were the lowest among the technologies assessed in the United Nations Economic Commission for Europe (UNECE) [Integrated Life-cycle Assessment of Electricity Sources report](#). The report noted that the emissions for nuclear power were front-loaded in the fuel chain.

The report’s model showed nuclear power in a range of 5.1–6.4 g CO₂ equivalent (eq.)/kWh, compared to a natural gas combined cycle plant (among the most efficient power generation systems, these are not found onboard ships) at 403–513 g CO₂ eq./kWh from a life cycle perspective, and between 92 and 220 g CO₂ eq./kWh with carbon capture and storage. Nuclear power, on average, has lower lifecycle GHG emissions than renewables such as solar or wind and is only rivalled by hydroelectric power.

Lifecycle emissions for power generation

Lifecycle GHG emissions, in g CO₂ equivalent (eq.) per kWh, regional variation, 2020



Source: <https://www.cameco.com/invest/markets/uranium-price>



3.2

Ship operator demand and interest

Interest in the use of nuclear power for shipping is driven by its emissions reductions. Nuclear power offers a path to the end goal of emissions reduction in shipping – zero emissions operation – and does so without the uncertainty of fuel and bunkering infrastructure development. The future availability of green fuel to operate a ship is no longer a concern once an agreement is in place for the lease of a reactor.

Recently, development projects have been initiated where shipping companies, shipyards and nuclear engineering companies are working in partnership on research and development of nuclear-propelled ship designs, including bulk carriers and containerships. LR is working across the industry on nuclear shipping projects and expects interest in nuclear propulsion will continue to increase as shipping emissions regulations tighten.

Client interest in nuclear propulsion is split between those looking to have assets in the water in the shortest time frame using current technologies – around 2030 – and those working on a longer horizon for deployment of around 20 years who will have a wider range of technologies to consider.

The main considerations for shipowners exploring nuclear power for ships are consistent with other alternative fuels – delivery timelines and cost. An additional factor for nuclear power is social acceptance of the use of nuclear reactors on commercial ships.

3.3

Techno-economic drivers

The comparison of the “cost of nuclear” is a fundamental question at the top of the agenda for those looking at maritime applications for nuclear power and is necessary for investment decisions. Before looking at how cost can be addressed, it is worth considering the likely deployment model for nuclear in maritime as this fundamentally impacts where capital expenditure (CAPEX) and operational expenditure (OPEX) land.

Firstly, it is reasonable to look at current onshore operating models and those being developed for new onshore applications such as heat and power to process plants e.g. Dow Corning’s Seadrift Plant in Texas. The Dow plant will be operated by the technology developer X-Energy, the primary reasons being X-Energy has the track record with the regulator and, unlike Dow Chemicals, is an expert in nuclear technology and operation.

Applying this approach to shipping would mean the reactor is likely to be owned and operated by a third party with a proven track record with nuclear regulators. Regulators will likely insist on a proven track record in the region of 10 years. The advantages to the maritime asset operator are twofold: They do not need to develop a comprehensive in-house nuclear capability and gain the approval of a nuclear regulator; and the reactor being owned by a separate entity means the CAPEX is not addressed up front by the asset owner.

For a ship to use the power of a reactor owned by a third party, the most likely scenario is a power by the hour arrangement where megawatt hours (MWh) are purchased by the ship operator from the reactor owner on a contracted basis. Opex is a well established cost as reactors are sealed units, hence total cost of life operation can be established with confidence before an asset is built. There may be

considerable residual value in the nuclear energy sources within a reactor at the end of its operational life; reactor owners will be equipped to redeploy these valuable materials to other assets

Compared to operation on traditional bunker fuels, issues around fuel price volatility are largely eliminated. Current work in this area has indicated that the cost of nuclear compares favourably with the cost of hydrocarbons today. The technology also poses very competitive new opportunities to operate assets in a different way, such as very fast steaming at negligible cost increase, which would in turn reduce the necessary fleet size. The cost and lifespan of a nuclear reactor may also lead to vessels with a design life closer to 50 years than the current 20-30 years; the same factors also make the likelihood of vessel retrofits low except in edge cases with very high value assets.

While the commercial model largely withdraws CAPEX cost from maritime asset operators’ concern, there is still an understandable desire to understand pricing. In this respect, a ball park figure of \$500M per reactor for those delivered in the early 2030’s, dropping off rapidly when production rates increase, is appropriate. It should be noted this is a simplistic example and does not include elements such as the regulatory journey for first-of-kind reactors.

Nuclear propulsion for container ships

Nuclear engineering company Core Power, which is developing nuclear technologies for the maritime sector, recently identified container ships over 10,000 twenty-foot equivalent unit (TEU) as prime candidates for adopting nuclear propulsion, particularly those serving the Asia-Europe trades due to EU ETS and Fuel EU Maritime savings.

Modelling the impact of introducing nuclear power into a typical Asia-Europe service, Core Power found that for a fleet of eight vessels, the average cost of each advanced reactor, along with its operation including insurance costs, cannot exceed \$3.8 bn over the course of 25 years to remain economically favourable compared to traditionally-fuelled vessels, a figure far above cost estimates for advanced reactors.

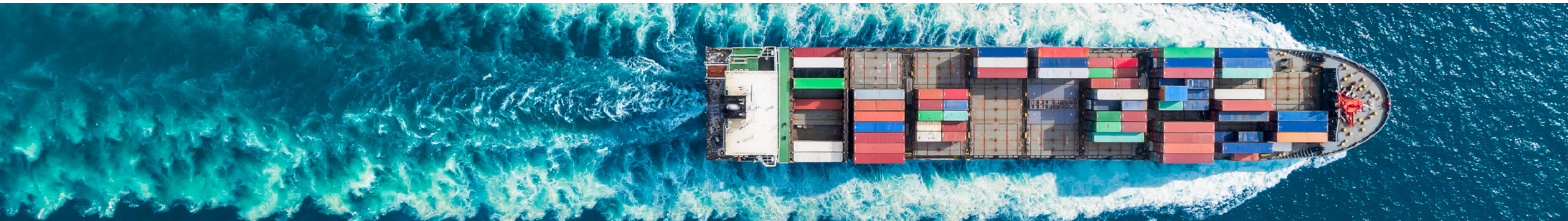
The study included capex and opex estimates for container ships based on the [Idaho National Laboratory's Configurations of Commercial Advanced Nuclear-Maritime Applications report](#), which puts capex for advanced reactors at between \$4,000 and \$7,000 per kWe.

70 MWe advanced reactor high and low capex estimates (\$m)

The table below shows high and low CAPEX and OPEX scenarios for a nuclear-powered container vessel on the Asia-Europe trade. The OPEX figures suppose replacing a ship's diesel engines with a nuclear reactor. The fuel consumption of a VLSFO vessel was converted to energy consumption, assuming 12MW-h per tonne of Very Low Sulphur Fuel Oil (VLSFO), and a \$15/MW-h to \$35/MW-h cost applied for the operation of an advanced reactor, based on Idaho National Laboratory's An Economics-by-Design Approach Applied to a Heat Pipe Microreactor Concept.

Containership capacity / TEU	CAPEX _{low} (\$m)	CAPEX _{high} (\$m)	Annual OPEX _{low} (\$m)	Annual OPEX _{high} (\$m)	25-year OPEX _{high} (\$m)
14,000	280	490	1.63	3.48	87
15,000	280	490	1.64	3.83	96
20,000	280	490	1.89	4.40	110
24,000	280	490	2.25	5.24	131

Source: Core Power based on Idaho National Laboratory estimates.



4.1

Chapter 4: Nuclear fuel production and supply

Introduction – overview

Fundamentally, nuclear fuel production and supply is not expected to be a concern for the shipowner. Shipowners and their operating partners will secure a guaranteed fuel supply for the duration of the reactor's lifespan through various structured contracting arrangements instead of treating fuel as an operational expense. This is a departure from other alternative fuels where the availability of fuels in sufficient quantities and necessary locations for bunkering a vessel persist throughout its lifespan.

The nuclear energy sources necessary for next generation reactors are currently available in quantities suitable for research reactors. Production capacities will need to increase significantly to meet commercial demand for small module reactor applications.

The High-assay low-enriched uranium (HALEU) fuels used by the new generation of reactor technologies are integral to their longevity and small physical footprint. The Low Enriched Uranium (LEU) sources used in current reactors for electricity generation are of between 3% and 5% enrichment. The HALEU fuels for future reactors are expected to fall between 10% and 20% enriched.



4.2

Supply and demand forecasts

There are an adequate number of producers of nuclear energy sources to create a competitive market and producers are expected to have the capacity to meet the forecast rising demand for nuclear fuel. Recent geopolitical tensions and the expected demand for HALEU have brought changes to government approaches to nuclear fuel production driven by energy security concerns and decarbonisation.

The US Department of Energy is pursuing multiple pathways to secure its own domestic supply of HALEU fuels for future reactor designs through down blending of government Highly enriched uranium (HEU) stocks and enrichment. The agency forecast demand of over 40 tonnes of HALEU by 2030.

The enrichment facility in Piketon, Ohio came online late in 2023 and was the first US-owned uranium enrichment plant to begin production since 1954. The site delivered 20kg of HALEU to the US Department of Energy in November 2023, and is expected to produce 900kg per year once at full capacity.

The 2022 US Inflation Reduction Act invested \$700m into the HALEU availability programme to address infrastructure and research gaps.

The [May 2022 HALEU report](#) by the EU’s Euratom Supply Agency (ESA) forecast HALEU demand of between 676 kg per year and 1256 kg per year in the EU by 2035. The report identified firm and meaningful HALEU quantity commitments as the single most important factor in enabling production increases.

EU producers of enrichment technology have claimed that a production facility for HALEU in the EU would only become commercially viable for demand of 3 to 8 tonnes per year. The bloc faces the decision of continuing to rely on US and Russian imports of HALEU, maintaining a 10-year stockpile of HALEU to ensure security of supply, or supporting an EU production facility. ESA estimates a six to seven year timescale to take a HALEU production facility from design to commissioning.

Uranium availability

The World Nuclear Association said in its biennial review of nuclear fuel demand and supply availability covering 2023-2024 that it had no doubt there are sufficient uranium reserves to meet future needs.

Kazakhstan is the main producer of uranium accounting for 43% of world production in 2022, followed by Canada at 14.9% and Namibia at 11.4%, according to figures from the World Nuclear Association.

Uranium production by country (tonnes U)

Region/country	Production 2022	Share in 2022 (%)	Change 2021/22 (%)	Production 2021	Share in 2021 (%)
Kazakhstan	21,227	43.0%	-2.7%	21,819	45.6%
Canada	7,351	14.9%	56.6%	4,693	9.8%
Namibia	5,613	11.4%	-2.4%	5,753	12.0%
Australia	4,553	9.2%	8.6%	4,192	8.8%
Uzbekistan	3,300	6.7%	-6.3%	3,520	74.4%
Russia	2,508	5.1%	-4.8%	2,635	5.5%
Niger	2,020	4.1%	-10.1%	2,248	4.7%
China	1,700	3.4%	6.3%	1,600	3.3%
Others	708	1.4%	1.9%	695	1.5%
South Africa	200	0.4%	4.2%	192	0.4%
Ukraine	100	0.2%	-78.0%	455	1.0%
United States	75	0.2%	837.5%	8	0.0%
Total	49,355	100.0%		47,810	100.0%

Source: Data from the WNA (August 2023) and specialised publications (because of rounding, tables may not add up)

4.3

Reactor decommissioning and radioactive waste

Addressing the public perception of radioactive waste and its management is necessary to obtain the social license to operate for nuclear power. All waste from nuclear power generation is tightly regulated and none is allowed to cause pollution. It is helpful to define the waste streams from a nuclear reactor in order to understand the regulations in place to protect human health and the environment.

While the decommissioning of nuclear reactors and subsequent disposal of radioactive material are expected to be the responsibility of the reactor owner rather than the shipowner, the subject will be of interest to those exploring the option of nuclear power in shipping.

Nuclear reactors create waste streams just as internal combustion engines and batteries create waste streams. However, the waste generated by nuclear reactors is unique in that it is small in volume, well-contained, and its radioactivity can be precisely measured. This allows for more controlled handling and disposal compared to waste from other technologies.

The International Atomic Energy Agency (IAEA) has established safety standards for the disposal of radioactive waste, which are adhered to by member countries to ensure the protection of human health and the environment.

Spent fuel is the most radioactive byproduct of operating a nuclear reactor. Handling of spent fuel is tightly regulated and carried out by specialists, and strict rules are in place to contain such material during transportation. Depending on its composition, spent fuel may be reprocessed to recycle its uranium and plutonium contents, ensuring the most efficient use of the material. Spent fuel is an example of a High-Level Waste (HLW), a category which accounts for 3% of the volume of produced radioactive waste and 95% of total radioactivity, according to the World Nuclear Association.

Low-Level Waste (LLW) makes up 90% of nuclear waste by volume and 1% by radioactivity, and includes items like clothing, tools, and rags that have become contaminated with radioactive substances during the operation and decommissioning of nuclear facilities. Some materials from designated active areas are marked as LLW as a precaution, even if they show no elevated levels of radioactivity. Certain advanced reactor designs may reduce LLW to near-zero levels.

Intermediate Level Waste (ILW) accounts for 7% of radioactive waste by volume and 4% by radioactivity. They differ from HLW in that they do not produce enough heat for it to be a factor in their containment and disposal. Examples include reactor components and fuel cladding.

Disposal of radioactive materials is defined as storage without any intention of retrieval, and regulations are in place governing the appropriate disposal of LLW, ILW, and HLW. LLW is commonly disposed of in near-surface sites where containers are placed in vaults at or close to ground level. Such sites are in operation in multiple European countries as well as Japan and the US. They are designed to handle materials with a half-life up to 30 years and may also handle short-lived ILW.

For the disposal of ILW and HLW, deep geological sites are preferred. These sites are designed to contain and isolate the radioactive waste from the environment for tens of thousands of years or longer, as the radioactivity of these materials decays over very long periods. HLW, in particular, is initially stored at the reactor site where it was produced, allowing its radioactivity to decrease to safer levels before it is transported to a final disposal site.

The management of radioactive waste from nuclear reactors is a highly regulated process that prioritises safety and environmental protection. With the potential for nuclear power to play a role in shipping, understanding the responsibilities and challenges associated with radioactive waste disposal is essential for stakeholders in the maritime industry.

5.1

Chapter 5:

Technology readiness

Introduction

Multiple competing reactor technologies are under development for nuclear power programmes and research in the field has been given more attention in recent years as energy security and decarbonisation have risen in political importance.

Six main technologies are recognised by the Generation IV International Forum (GIF), a group with the goal of coordinating international cooperation on the development of advanced reactor technologies, representing 13 countries and the EU.

For shipping, the most promising technologies include refinements of existing Pressurised Water Reactor (PWR) designs and certain of the less proven Generation IV technologies which are in various stages of development. For all the technologies, the main challenges are the setting of standards for their use in shipping, and proving the operational safety of reactor designs.

For maritime applications, the most promising reactors with the shortest time to market are the following:

- Pressurised water reactors (PWR)
- Heat pipe micro reactors (HPMR)
- Molten salt reactors (MSR)
- Lead cooled fast reactors (LFR)
- High temperature gas reactors (HTGRs)



5.2

Pressurised water reactors

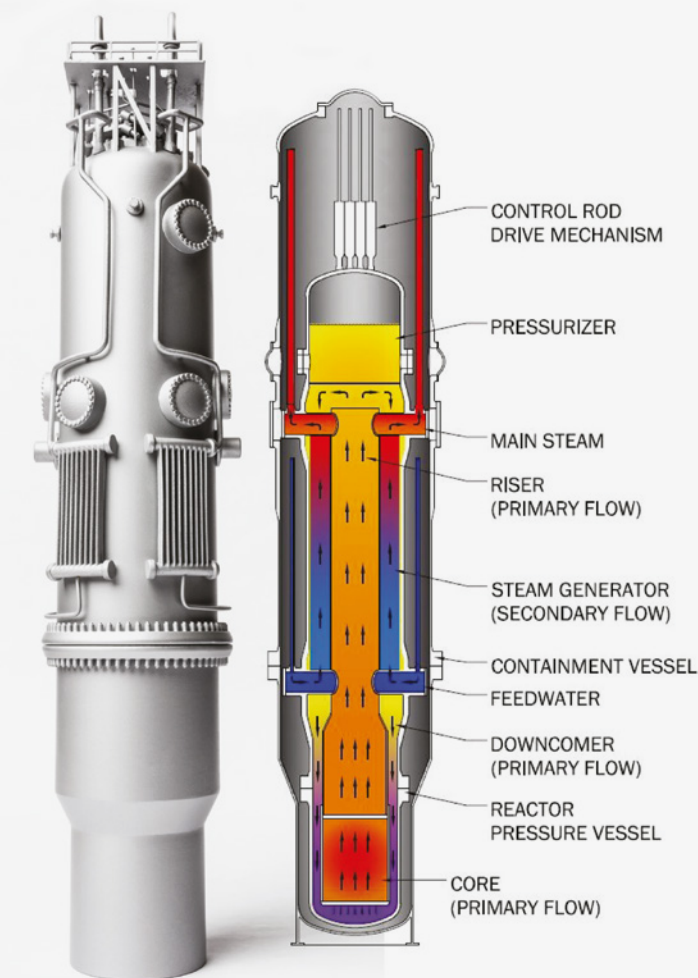
PWRs have been developed into small modular reactors that could be used onboard. They use uranium as fuel to generate heat through a controlled nuclear fission chain reaction. The heat is transferred to the coolant, which in a PWR is typically ordinary water. The heated coolant remains in a liquid state at the high temperatures due to the high pressure maintained within the reactor vessel. This heat can then be used for conversion to electrical power, mechanical power, or direct thermal energy for heating purposes.

For maritime applications, the development of smaller PWR designs suitable for factory fabrication with passive safety features brings the prospect of commercial deployment of PWR.

PWRs are unique among the reactor technologies in that there is experience of operating such reactors in a marine environment through multiple militaries, and a few historic examples of government-backed merchant vessels.

For production of a small modular PWR – the type necessary for broader adoption in the maritime sector – development is at a similar level to other technologies where validation of an integrated prototype is underway in a test environment.

Diagram of a pressurised water reactor



5.3

Heat pipe micro reactors

A nuclear heat pipe (low pressure microreactor) is a passive heat transfer device. Uranium is used as fuel to generate heat through a controlled nuclear fission chain reaction in the reactor core. The heat is transferred to a working fluid in the heat pipe. This heat can then be used for conversion to electrical power, mechanical power, or direct thermal energy for heating purposes.

The passive nature of heat pipes means no coolant is required, eliminating the need for associated pumping systems. Designs include multiple passive safety systems to ensure the removal of heat in the event of a malfunction. The impact of a potential failure is reduced by the low pressure nature of the reactor, reducing the kinetic energy on hand to disperse materials.

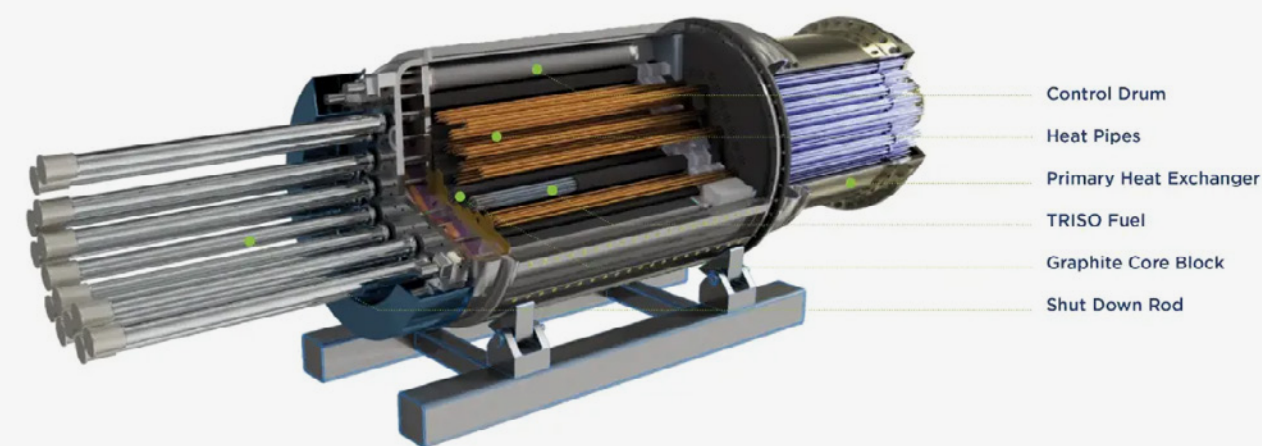
Heat pipe reactors designs include remote monitoring capabilities; there will be no need for any onboard operations.

Once the fuel in a heat pipe reactor is depleted, the reactor will be removed from the vessel and replaced with a new unit. The depleted reactor will be taken away for processing outside of the port environment in a specialist facility. Replacement in the lifetime of the vessel may not be required.

The TRi-structural ISOtropic particle fuel (TRISO) fuel used in heat pipe and gas cooled reactors exists at demonstration level, but there is no large-scale production.

Technology demonstration projects of heat pipe micro reactors are ongoing, including at BWXT and Westinghouse, for both defence and civil applications

Diagram of a heat pipe micro reactor



The reactivity control drums are used to moderate the power output of eVinci such as for load following applications or shut down. Otherwise, they are stationary.

Westinghouse's eVinci™ Microreactor. Image courtesy of Westinghouse

5.4

Molten salt reactors (MSR)

A molten salt (low-pressure) reactor uses uranium dissolved in a molten fluoride salt liquid mixture to generate heat through a controlled nuclear fission chain reaction in the core. The molten salt acts as a coolant and a moderator, sustaining the chain reaction. The heat is transferred to a working fluid. This heat can then be used for conversion to electrical power or direct thermal energy for heating purposes. Some MSR designs allow for online refuelling without shutting down the reactor.

Certain MSR designs have the potential to use spent fuel from light water reactors as a fuel.

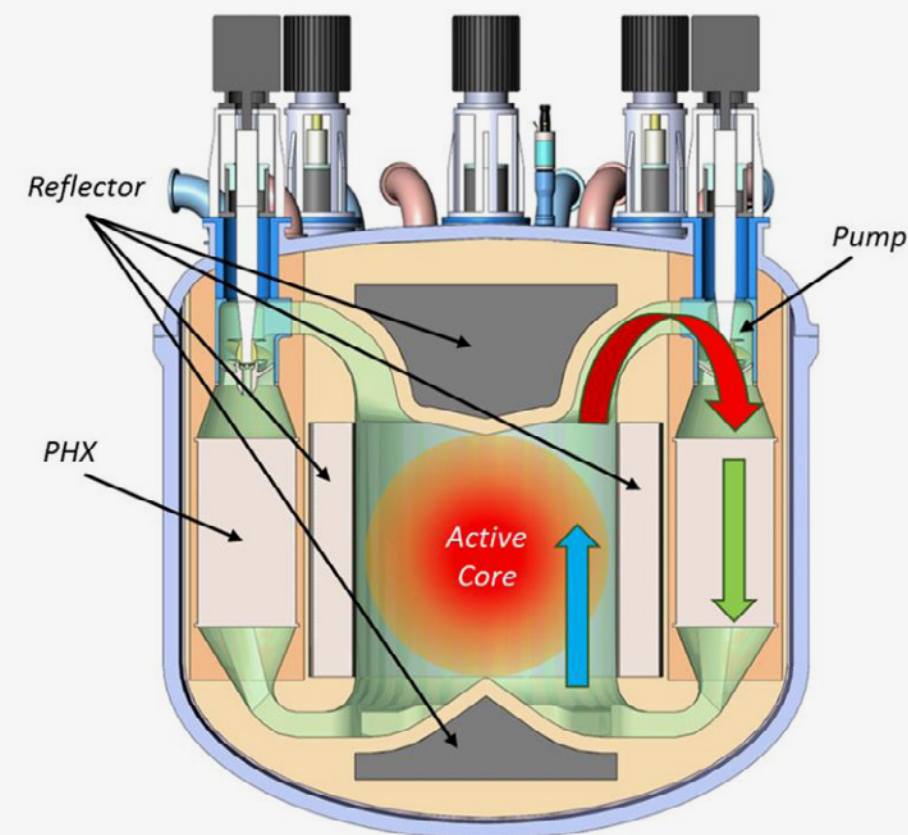
Experimental test facilities exist, both full scale and separate effect test facilities for various MSR technologies. The technology is undergoing validation and will require marinisation.

Therefore MSRs fuel exists at an experimental level, but there is no large scale demonstration of fuel production.

MSRs have multiple inherent passive safety features. Properties of the liquid fuel make it prone to cooling in the event of overheating or containment breach, and passive safety devices in the reactor drain the fuel into a tank for cooling in the event of malfunction.

Their low pressure design reduces material dispersal in a containment breach and the fuel salts are inert, removing safety concerns over chemical reactions.

Diagram of a molten salt reactor



TerraPower's molten chloride fast reactor. Image courtesy of TerraPower

5.5

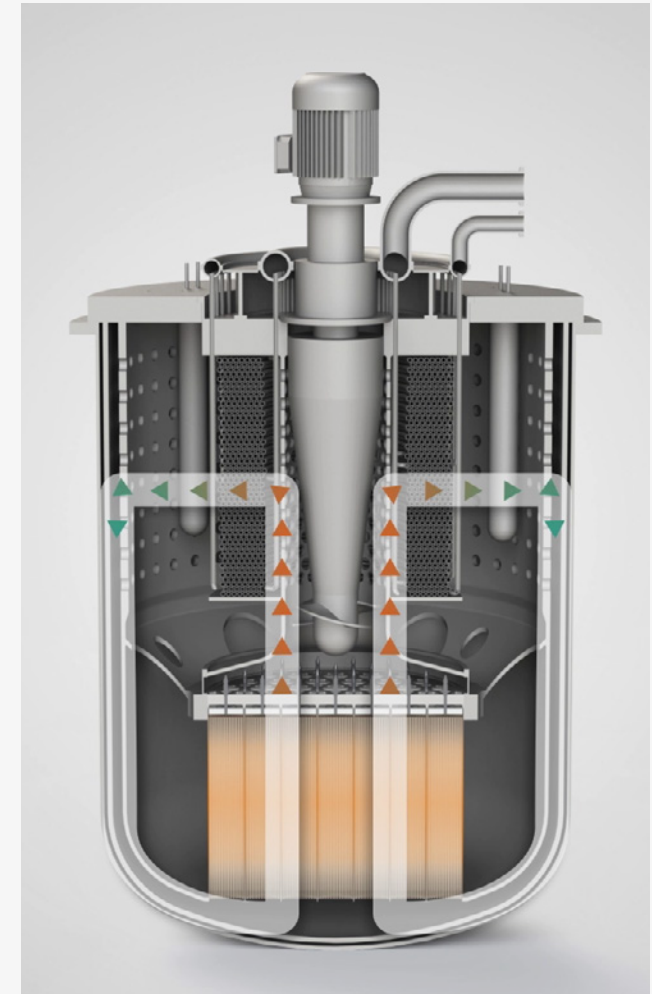
Lead-cooled fast reactors (LFRs)

Lead-cooled fast reactors (LFRs) use depleted uranium, plutonium and minor actinides as fuel to generate heat through a controlled nuclear fission chain reaction in the core. Lead is used as a coolant to remove heat from the core and transfer it to water to generate steam for energy generation.

Passive safety features reduce the core temperature when it rises above normal operating levels without operator intervention.

The fuel and technology exist and has been tested in maritime through defence applications, including through reactor maintenance and refuelling in defence applications.

Diagram of lead-cooled fast reactor



NewCleo's TL-30 LFR. Image courtesy of NewCleo

5.6

High-temperature gas reactors

High temperature gas reactors (HTGRs) have been developed using a range of fuels from low to high enriched uranium, thorium, and plutonium. The reactors use ceramic coated pellets as a fuel to generate heat through a controlled nuclear fission chain reaction in the reactor core. Helium as a coolant to extract heat for conversion to electrical power or direct thermal energy for heating purposes. The technology has applications in providing process heat and in hydrogen production.

TRISO reactor fuel for HTGRs exists in demonstration quantities but no large scale production is available. The reactor technology is operational on land, but is not proven in maritime application.

The abundant use of graphite in the core slows any temperature fluctuations during malfunctions. The inert nature of helium and the thermal resistance of the ceramic fuel pellets enable passive heat removal in the event of an accident without operator intervention.

Diagram of a high temperature reactor

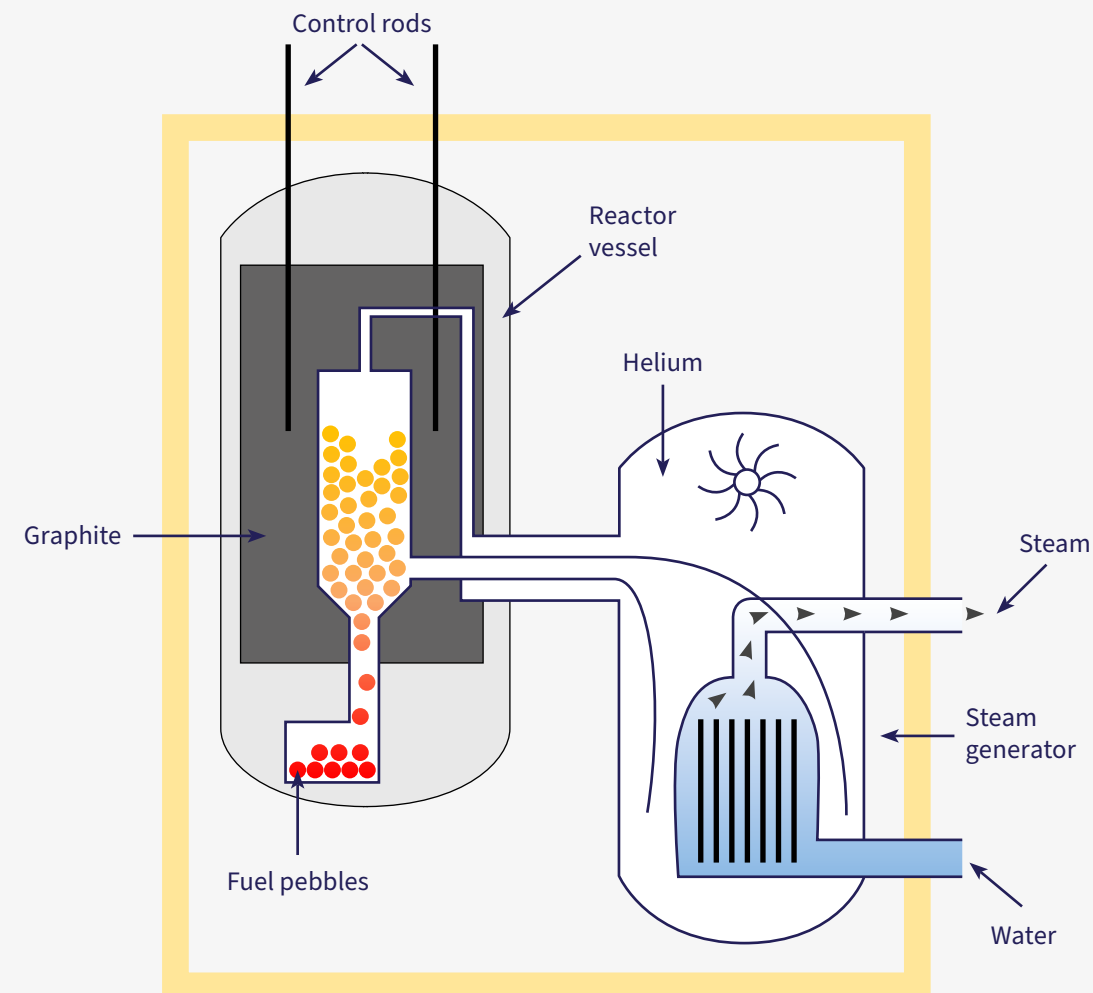


Diagram of a high temperature reactor - Image courtesy of World Nuclear Association.

Chapter 6

Summary and conclusion

LR will continue to follow developments in nuclear shipping closely and cover them in future updates to this guide.

Nuclear power can be seen as a transformational technology rather than an incremental or even step change improvement. It will not be a direct replacement for oil-fired systems like some alternative fuels, but rather a primary catalyst to fundamentally reshape the shipping industry. Part of the transformation would be safer, more reliable, emissions free, longer-lived, and more productive ships. Another part of the transformation will be in the ship operator's structure, including technical management, procurement, approach to quality, and an elevated safety culture.

The social licence to operate and regulatory ambiguity are the primary challenges for the early adoption of small modular reactors and their application in the maritime industry. National regulations and approvals on a case-by-case basis are expected in the near term until global standards are agreed at the IMO and IAEA.

Nuclear power holds immense potential for revolutionising the maritime industry, offering a path towards sustainable and efficient shipping solutions. While its application in naval operations has a proven track record, the widespread adoption of nuclear power in commercial shipping is on the horizon, driven by advancements in technology and a growing recognition of its benefits.

The journey towards commercially viable nuclear-powered shipping is marked by collaboration and innovation. Governments worldwide are actively supporting the development of small modular reactors (SMRs) for power generation, fostering a dynamic market of private companies pioneering cutting-edge technologies. Building upon decades of accumulated knowledge, these SMRs represent a leap forward in reactor design, emphasising safety, efficiency, and modularity for streamlined production.

Regulatory frameworks are evolving to accommodate the unique characteristics of SMRs and their application in maritime settings. This evolution involves a shift towards goal-based regulations that encourage active engagement from designers, builders, and operators, fostering a culture of safety and responsibility. International organisations like the IMO and IAEA are working towards establishing global standards, paving the way for wider adoption.

The safety of nuclear power in shipping remains paramount. Stringent safety protocols and advanced reactor designs prioritise the protection of both crew and environment. As SMR technology matures and regulatory clarity increases, ship designs optimised for nuclear propulsion will emerge, ushering in a new era of efficient and environmentally friendly vessels.

The collective efforts to take advantage of these opportunities and advance nuclear-powered commercial shipping underscore the immense value of this technology. With its potential to provide a safe, zero-emission fuel source that eliminates the need for frequent refuelling, nuclear power promises to transform the shipping industry, contributing to a cleaner and more sustainable future for global trade and transportation.

7.1

Chapter 7

Other resources and annexes

Links and other resources

Safety

[IAEA Safety Standards – Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards](#)

[IAEA Safety Standards – Disposal of Radioactive Waste](#)

Commercial

[Cameco uranium price averages](#)

[World Nuclear Association World Uranium Mining Production data](#)

[EU carbon pricing brings new pressures – and new plays – to maritime](#)

[UNECE – Carbon Neutrality in the UNECE Region: Integrated Life-cycle Assessment of Electricity Sources](#)

[ESA – HALEU report May 2022](#)

Regulatory

[Lloyd's Register – What is the EU ETS and EU MRV?](#)

[European Commission – The Fuel EU Maritime Regulation](#)

Annex 1: Technology, Investment and Community readiness levels (TRL, IRL, CRL) and definitions

There are three readiness levels used in this report: technology, investment and community. All are on a scale, with TRL on a scale of one to nine, and CRL and IRL on a scale of one to six.

Technology readiness (TRL)

The technology readiness level indicates the maturity of a solution within the research spectrum from the conceptual stage to being marine application ready. It is based on the established model used by NASA and other agencies and institutes, using a nine-level scale.

Level	Technology Readiness Level (TRL)	
1	Idea	Basic principle observed
2	Concept	Technology concept formulated
3	Feasibility	First assessment feasibility concept and technologies
4	Validation	Validation of integrated prototype in test environment
5	Prototype	Testing prototype in user environment
6	Product	Pre-production product
7	Pilot	Low-scale pilot production demonstrated
8	Market introduction	Manufacturing fully tested, validated and qualified
9	Market growth	Production and product fully operational

Investment readiness level (IRL)

The investment readiness level indicates the commercial maturity of a marine solution on the spectrum from the initial business idea through to reliable financial investment. It addresses all the parameters required for commercial success, based on work by the Australian Renewable Energy Agency (ARENA). The six-level scale used summarises the commercial status of the solution and is determined by the available evidence in the market.

INVESTMENT READINESS LEVEL (IRL)		
1	Idea	Hypothetical commercial proposition
2	Trial	Small-scale commercial trial
3	Scale up	Commercial scale up
4	Adoption	Multiple commercial applications
5	Growth	Market competition driving widespread development
6	Bankable asset	Bankable asset class

More details on the readiness levels adopted by Lloyd's Register can be found on the [LR Maritime Decarbonisation Hub](#) zero carbon fuel monitor.

Community readiness level (CRL)

The community readiness level indicates the societal maturity of a marine solution in terms of acceptability and adoption by both people and organisations. It is gauged on the spectrum from societal challenge through to widespread adoption. CRL is based on the work by ARENA and Innovation Fund Denmark adapted to a six-level scale.

COMMUNITY READINESS LEVEL (CRL)		
1	Challenge	Identifying problems and expected societal readiness, formulation of possible solution(s) and potential impact
2	Testing	Initial testing of proposed solution(s) together with relevant stakeholders
3	Validation	Proposed solution(s) validated, now by relevant stakeholders in the area
4	Piloting	Solution(s) demonstrated in relevant environment and in cooperation with relevant stakeholders to gain initial feedback on potential impact
5	Planning	Proposed solution(s) as well as a plan for societal adaptation completed and qualified
6	Proven solution	Actual project solution(s) proven in relevant environment

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